

title: E · International Benchmarks description: Comparable government HRIS systems worldwide — patterns worth borrowing, disasters worth avoiding, and one uncovered value-add candidate from Chile.

E · International Benchmarks

!!! info “How this paper is used” This paper does **not** propose new scope. It documents comparable government HRIS systems worldwide and extracts patterns worth borrowing (and disasters worth explicitly avoiding). Every reference has a direct implication for one of Papers **A / B / C / D**.

- ****Cite in Paper B §5 (Architecture)** and **§15 (Risk)** to show the design has international p**
- ****Anchor Paper D's value-added components** in observed practice.**
- ****Arm the oral-defence team** with concrete phrases evaluators recognise.**

Selection criteria

Only systems that meet **all four**:

1. Government HRIS at **≥ 100,000 employee scale**
2. **Deployed in production** (not proof-of-concept, not a white paper)
3. Public-sector — ideally education, otherwise a directly adjacent national HR/payroll system
4. **English-accessible** documentation from a primary source (government, World Bank, OECD, or vendor case study)

Education-department analogues (closest fit)

System	Country	Scale	Signature idea worth borrowing	Applies to
Shala Darpan + UDISE+	India	~9 M teachers, 1.5 M schools	SMS-first teacher self-service, Aadhaar-linked eKYC deduplication of 201 file records	D §1
DIKSHA	India	4 M+ teachers on the platform	Training-completion feeds directly into appraisal and career progression — no separate LMS integration project	D §14 (Batch 3)
TSC + TPAD	Kenya	~340 K teachers	Mobile-first Teacher Performance Appraisal, purpose-built for field connectivity	D §2
PERSAL / PERSOL / PILIR	South Africa	1.2 M civil servants (incl. teachers)	Longest-running large-scale public payroll on the continent; battle-tested integrations to pensions/medical aid (analogue to GSIS/PhilHealth)	A · C
SNED + SIGE	Chile	~240 K teachers	Constraint-programming solver that suggests teacher-to-school placements against distance, subject, seniority	new VAC (see below)
Novopay (post-2015 rebuild)	New Zealand	~110 K teachers	The rebuild after the 2012 disaster is the reference model for how to do payroll parallel-runs safely at scale	A M6 · B §9

National public-sector at scale (adjacent inspiration)

System	Country	Scale	Signature idea worth borrowing	Applies to
NHS Electronic Staff Record (ESR)	UK	1.5 M NHS staff	Employee self-service + learning-management fully integrated — reference for “one login, one profile” at scale	C · D §14
HRMIS	Malaysia	1.6 M civil servants	30+ modules on a strict BPMN workflow engine — the pattern the PBD implicitly asks for	C
HRP@Gov / OneCV	Singapore	~150 K public servants	Competency framework as a first-class primitive; career pathways <i>published</i> to employees	A · D §10 (Batch 1)
X-Road / eesti.ee	Estonia	~30 K public servants (whole-of-gov)	Cross-agency data exchange with a mandatory per-query audit log — direct precedent for D §5 hash-chain ledger	D §5
e-People	South Korea	~1.1 M civil servants	Predictive HR analytics dashboards for management, at ministry scale	D §3 · D §9 (Batch 2)
National Finance Center (NFC)	USA (federal)	~600 K federal employees	Shared-service payroll model — a template for how DepEd’s HRIS could later serve other agencies	D §7
SAPK / SIASN	Indonesia	~4.2 M ASN (civil servants)	Continental-scale, multi-datacentre deployment with delegated data ownership per ministry	C (deployment view)
PSB HR portal	Georgia (country)	~130 K public servants	Modern citizen-facing transparency APIs; every civil-servant appointment is publicly discoverable	D §6

Six patterns worth borrowing (and citing by name)

1. India · Aadhaar-linked eKYC → PhilSys

Aadhaar linkage let India dedupe millions of ghost records overnight. PhilSys linkage lets DepEd do the same. Already **Paper D Batch 3 #13** (PhilSys SSO + digital signature).

2. Kenya · Mobile-first TPAD

Proves that field-focused mobile apps beat desktop web for the last mile. Reinforces **Paper D §2** (Offline PWA + CRDT sync).

3. Chile · Teacher placement optimisation

A CP-SAT solver that recommends teacher-to-school assignments against distance, subject, seniority, and hardship-post premiums. This is a **completely new VAC candidate** — see §Uncovered VAC below.

4. NHS ESR · SSO + LMS integration

Self-service integrated with learning management at 1.5 M-staff scale. Fits with **Paper D §14 (Batch 3)** (micro-learning nudges).

5. Malaysia · Workflow-first design (BPMN engine)

Validates the base scope choice in Paper C. Cite in **Paper B §5 Architecture** as prior art — 1.6 M civil servants on a workflow-driven HRIS proves the pattern scales.

6. Estonia · Per-query audit log via X-Road

Validates the hash-chained audit ledger in **Paper D §5**. Cite in Paper B §7 (Security) as international precedent for the design.

Three disasters we've studied and avoid

Naming these in **Paper B §15 Risk Management** signals maturity — evaluators will recognise them.

Novopay teacher payroll — New Zealand, 2012

What went wrong. Talent2 (later Deloitte) launched the new payroll without a proper parallel-run alongside the legacy system. Teachers were underpaid, overpaid, or unpaid for months. Estimated remediation cost NZ\$45 M+. Cabinet-level scandal; ministerial resignations.

How we avoid it. **Paper D §4** (payroll anomaly detector at the M6 parallel-run gate). The PBD-mandated M6 parallel run is already the safety net; the anomaly detector *hardens* it by refusing to sign off if net-pay deltas, ghost-employee patterns, or remittance drift exceed thresholds set by DepEd finance.

Queensland Health payroll — Australia, 2010

What went wrong. IBM-led implementation of SAP payroll went live without adequate fallback; 78,000 health workers had payslip errors for years. Scope explosion from 315 rules to 24,000+ pay rules mid-project. AU\$1.2 B+ overrun. IBM was sued and lost.

How we avoid it. **Paper B §9** milestone entry/exit criteria (any milestone can trigger a controlled rollback to the previous state) + **Paper D §18 (Batch 4)** (outcome-based sleeve) — the bidder has explicit skin in the game against a re-run of this failure pattern.

OPM breach — USA, 2015

What went wrong. US Office of Personnel Management's HRIS was breached; **21.5 million SF-86 clearance records** exfiltrated, including biometrics. Root cause: legacy stack, no MFA on admin endpoints, insufficient network segmentation, no tamper-evident logging.

How we avoid it. **Paper D §5** (hash-chained audit ledger — tampering becomes provable, not just detectable) + **Paper B §7 Security** (ABAC + row-level security + encryption at rest, MFA on every administrative surface) + regular red-team reviews (Paper D §11, Batch 2).

Uncovered VAC candidate — Chile-style teacher placement optimiser

Update (post-review): this candidate has been **promoted to Paper D as VAC #8** (advisory teacher-to-school placement optimiser). See [Paper D §8](#) for the design and delivery plan, and [Paper B §16.5](#) for the contractual wording.

Original analysis retained below for context:

Teacher-to-school placement optimiser — a small optimisation service (Google OR-Tools CP-SAT) that ingests plantilla vacancies, subject requirements, current teacher assignments, distance-to-school, seniority tenure, and hardship-post premiums. It emits candidate reassignments and a Pareto-front of trade-offs (e.g. “reassign 12 teachers to close 8 vacancies at the cost of 47 km average commute increase”). DepEd Central approves, rejects, or overrides each recommendation; decisions feed back as constraints for the next cycle.

Why it belongs in Paper D.

- Directly addresses the perennial DepEd headline “*no math teachers in Mindanao*”.
- Turns the plantilla module from a record store into a **management decision tool**.
- No competitor for this bid is likely to offer this.
- Marginal delivery cost is small: existing open-source solver + a thin service on the base infra.
- Milestone slot: M6 (data model + first solver runs), M7 (UI + decision workflow), M8 (production hand-off).

Where it plugs in. Uses the plantilla + 201 file data already in Paper C’s Recruitment and Core-HR contexts; reuses the workflow engine and approval routing from the base scope.

Cross-reference matrix

International pattern	Paper A	Paper B	Paper C	Paper D
India Aadhaar eKYC	✓		Auth/identity	\$13 (Batch 3)
Kenya mobile-first TPAD	✓		Mobile/PWA	\$2
Chile placement solver			Recruitment	proposed #23
NHS ESR SSO + LMS			Component view	\$14 (Batch 3)
Malaysia BPMN workflow-first		\$5 Architecture	Container view	
Estonia per-query audit		\$7 Security	Audit schema	\$5
Novopay lesson	M6	\$9 Milestones · \$15 Risk		\$4
Queensland Health lesson		\$9 · \$15 Risk		\$18 (Batch 4)
OPM breach lesson		\$7 Security · \$15 Risk		\$5 · \$11 (Batch 2)
Singapore HRP@Gov competency			Career/Succession	\$10 (Batch 1)
Korea e-People analytics			Analytics tier	\$3 · \$9 (Batch 2)
USA NFC shared-service payroll		\$11 Continuity		\$7
Georgia PSB transparency APIs				\$6

Sources & further reading

Every claim above is traceable to a primary source. Consolidated bibliography for the annex:

- **India** — MoE Shala Darpan portal (shaladarpan.education.gov.in), UDISE+ annual reports, DIKSHA impact assessments (Central Square Foundation, 2022–2024).
- **Kenya** — TSC annual reports, TPAD system documentation on tsc.go.ke , World Bank *Kenya Teacher Management Project* completion report.
- **South Africa** — DPSA PERSAL modernisation reports, National Treasury *Public Service Wage Bill Reviews*.
- **Chile** — MinEduc SIGE technical documentation, OECD *Education Policy Review: Chile* (2017, 2023).
- **New Zealand** — *Novopay Review* (Ministerial Inquiry, 2013), *Education Payroll Ltd* annual reports 2016–2024.
- **United Kingdom** — NHS ESR system documentation on my.esr.nhs.uk , National Audit Office reports.
- **Malaysia** — JPA HRMIS technical documentation, MAMPU digital-government roadmap.
- **Singapore** — GovTech blog on OneCV / HRP@Gov, PSD annual reports.
- **Estonia** — RIA X-Road documentation, e-Estonia case studies (e-estonia.com).
- **South Korea** — Ministry of Personnel Management annual reports, KIPA papers on e-People.
- **United States** — USDA NFC documentation, OMB *Federal Human Resources Line of Business*, OPM breach post-mortems (GAO GAO-16-88, US House Oversight report).
- **Indonesia** — BKN SIASN documentation, MENPAN-RB SPBE annual reviews.
- **Georgia** — Public Service Bureau *Transparency Reports*, EU4Digital case studies.

Where available, permalinks are recorded in the repository at [references/E_sources.md](#) (post-award).

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